

BSDSF23-Morning

Q1: Four married couples have bought 8 seats in the same row for a concert. In how many different ways can they be seated if all the men sit together to the right of all the women?

Theorem 2.1: The number of permutations of n objects is $n!$.

the members of each gender can be seated in $4!$ ways. Then using Theorem 2.1, both men and women can be seated in $(4!)(4!) = 576$ ways.

Q2: In how many ways can a caravan of 8 covered wagons from Arizona be arranged in a circle?

Theorem 2.3: The number of permutations of n objects arranged in a circle is $(n - 1)!$.

By Theorem 2.3, there are $7! = 5040$ arrangements.

Q3: How many ways are there to select 3 candidates from 8 equally qualified recent graduates for openings in an accounting firm?

Theorem 2.6: The number of combinations of n distinct objects taken r at a time is

By Theorem 2.6, there are $\binom{8}{3} = 56$ ways.

Q4: In testing a certain kind of truck tire over rugged terrain, it is found that 25% of the trucks fail to complete the test run without a blowout. Of the next 15 trucks tested, find the probability that For $n = 15$ and $p = 0.25$, we have

(a) from 3 to 6 have blowouts

$$(a) P(3 \leq X \leq 6) = P(X \leq 6) - P(X \leq 2) = 0.9434 - 0.2361 = 0.7073.$$

(c) more than 5 have blowouts.

$$(c) P(X > 5) = 1 - P(X \leq 5) = 1 - 0.8516 = 0.1484.$$

Q5: From a lot of 10 missiles, 4 are selected at random and fired. If the lot contains 3 defective missiles that will not fire, what is the probability that [7.5 + 7.5 = 15]

(a) all 4 will fire?

$$(a) \text{Probability that all 4 fire} = h(4; 10, 4, 7) = \frac{1}{6}$$

(b) at most 2 will not fire?

$$(b) \text{Probability that at most 2 will not fire} = \sum_{x=0}^2 h(x; 10, 4, 3) = \frac{29}{30}$$